

WEYL-CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL-RIESZ TRANSFORM COMMUTATORS

ZHIJIE FAN, JI LI, AND DMITRIY ZANIN

ABSTRACT.

CONTENTS

1. Introduction	1
2. Preliminaries	2
2.1. Harmonic analysis in the Dunkl setting	2
2.2. Weyl chamber as simplicial cone	3
2.3. Nearly weakly orthonormal sequences	3
2.4. Principal symbol and crossed products	4
2.5. Birman-Solomyak lemmas	5
3. Chamber is a doubling metric measure space	5
4. Rochberg-Semmes based estimate	7
5. Proof of special Cwikel estimate in Dunkl setting	11
6. Proof of Theorem 1.1 (i)	14
7. Proof of Theorem 1.1 (ii)	16
7.1. Abstract lower estimate	16
7.2. Proof of Theorem 1.1 (ii)	19
8. Proof of smoothness	20
9. Approximation theorem	25
9.1. First approximation theorem	25
9.2. Second approximation theorem	27
9.3. Proof of Theorem 9.1	29
10. Spectral asymptotic formula	32
References	35

1. INTRODUCTION

Theorem 1.1. *Let R be a normalised root system in $\mathbb{R}^N$, $N \geq 2$, and let G be the corresponding finite reflection group. Let $\kappa : R \rightarrow \mathbb{R}_+$ be G -invariant multiplicity function. Let $R_{\text{Dunkl},j}$ be the j -th Dunkl-Riesz transform corresponding to R and κ . Let w be the Dunkl weight corresponding to R and κ . Let $f \in L_1^{\text{loc}}(\mathbb{R}^N)$.*

2010 *Mathematics Subject Classification.* 47B10, 42B20, 43A85.

Key words and phrases. Schatten class, Riesz transform commutators, Dunkl operator, Besov space, Nearly weakly orthogonal sequence.

- (i) If $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, then $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_\kappa \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

- (ii) If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, then $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \geq c_\kappa^{-1} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

- (iii) For every $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ and for every $1 \leq j \leq N$, there exists a limit

$$\lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \text{write expression}.$$

2. PRELIMINARIES

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $|\cdot|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane orthogonal to a normalised vector v is defined by

$$\sigma_v x := x - 2\langle x, v \rangle v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called normalised if $|v|^2 = 1$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R .

Let $\kappa : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper.

Define $d(x, y) = \min_{g \in G} |x - gy|$. Recall from [8, Chapter VII, proof of Theorem 2.12] that $d(x, y) = |x - y|$ iff x and y belong to the same chamber.

Define the Dunkl weight on $\mathbb{R}^N$ by

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{\kappa(v)}, \quad x \in \mathbb{R}^N.$$

Dunkl measure $w(x)dx$ is also denoted by w .

Let $B(x, r) := \{y \in \mathbb{R}^N : |x - y| < r\}$. Let $\mathbf{N} := N + \sum_{v \in R} \kappa(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, \ t, r > 0.$$

Furthermore,

$$(1) \quad w(B(x, r)) \sim r^N \prod_{v \in R} (|\langle x, v \rangle| + r)^{\kappa(v)},$$

where the implicit constant only depends on N, R and κ . It follows from (1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^N \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{\kappa(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} = - \sum_{j=1}^N T_{e_j}^2.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} = T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}$, $j = 1, 2, \dots, N$.

Operators $U_g : f \rightarrow f \circ g^{-1}$, $g \in G$, are unitary both in $L_2(\mathbb{R}^N)$ and in $L_2(\mathbb{R}^N, w(x)dx)$. Obviously, $g \rightarrow U_g$ is a unitary representation of the group G . To lighten the notations, we write U_v instead of U_{σ_v} for $v \in R$.

2.2. Weyl chamber as simplicial cone.

Definition 2.1. *A connected component of the set*

$$\{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber.

The following assertion can be found in e.g. Theorem 2 on p.166 in [1].

Theorem 2.2. *Let Γ be a chamber. There exists a linear isomorphism $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ such that $\Phi(\Gamma) = R_+^{N_1} \times \mathbb{R}^{N_2}$ (with $N_1 + N_2 = N$).*

2.3. Nearly weakly orthonormal sequences. Rochberg and Semmes [12] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [9, 11, 3, 4, 10, 12, 14, 13]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [9, Section 13] for a complete proof).

Notation 2.3. *Let*

$$\mathbb{D} = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube in $\mathbb{D}_k$ is of the shape

$$\frac{m}{2^k} + [0, 2^{-k})^N, \quad m \in \mathbb{Z}^N.$$

For every dyadic cube $Q \in \mathbb{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

Definition 2.4 and **Lemma 2.5** are stated as my understanding of Hytonen's Section 7. Hytonen DOES NOT STATE HIS ASSUMPTIONS ON MEASURE!!!!

Definition 2.4. *Let measure $\mathbf{w}(x)dx$ be doubling. Let $c > 0$ and $2 < r \leq \infty$. A sequence of functions $\{e_Q\}_{Q \in \mathbb{D}}$ is called (r, c) -NWO if*

- (i) $\text{supp}(e_Q) \subseteq cQ$ for every $Q \in \mathbb{D}$;
- (ii) $\|e_Q\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} \leq \mathbf{w}(Q)^{\frac{1}{r} - \frac{1}{2}}$ for every $Q \in \mathbb{D}$;

Lemma 2.5. *Let measure $\mathbf{w}(x)dx$ be doubling. Let $0 < p < \infty$. Suppose that A is an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\begin{aligned} \|A\|_{\mathcal{L}_{p,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))} &\leq C_{w,r_1,c_1,r_2,c_2,p} \|\{\lambda_Q\}_{Q \in \mathbb{D}}\|_{p,\infty}, \\ \text{dist}_{\mathcal{L}_{p,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))} \left(A, (\mathcal{L}_{p,\infty})_0(L_2(\mathbb{R}^N, \mathbf{w}(x)dx)) \right) &\leq \\ &\leq C_{\mathbf{w},r_1,c_1,r_2,c_2,p} \text{dist}_{l_{p,\infty}(\mathbb{D})} \left(\{\lambda_Q\}_{Q \in \mathbb{D}}, (l_{p,\infty})_0(\mathbb{D}) \right). \end{aligned}$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [9]. $\square$

Lemma 2.6. *Let measure $\mathbf{w}(x)dx$ be doubling. Let $1 < p < \infty$. Suppose that A is an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\left\| \left\{ \langle A e_Q, f_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{p,q}(\mathbb{D})} \leq c_{\mathbf{w},r_1,c_1,r_2,c_2,p,q} \|A\|_{\mathcal{L}_{p,q}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}.$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [9]. $\square$

2.4. Principal symbol and crossed products. Let Π be the C^* -subalgebra in $B(L_2(\mathbb{R}^N))$ generated by M_f , $f \in (C_0 + \mathbb{C})(\mathbb{R}^N)$ and $h(\frac{\nabla}{\sqrt{\Delta}})$, $h \in C(\mathbb{S}^{N-1})$. Note that every compact operator is contained in Π . There exists a surjective $*$ -homomorphism

$$\text{sym} : \Pi \rightarrow (C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})$$

such that $\ker(\text{sym})$ consists exactly of all compact operators.

We have a natural action α of G on $B(L_2(\mathbb{R}^N))$ and, therefore, on Π — the C^* -algebra of operators admitting principal symbol. Namely,

$$\alpha(g) : T \rightarrow U_g T U_g^{-1}, \quad T \in B(L_2(\mathbb{R}^N)), \quad g \in G.$$

Define the action β of G on $(C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})$ by setting

$$(\beta(g))(f \otimes h) = (f \circ g^{-1}) \otimes (h \circ g^{-1}).$$

The same action extends to von Neumann algebra $L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})$. We have an equivariance relation

$$\text{sym}(\alpha(g)(T)) = (\beta(g))(\text{sym}(T)).$$

Clearly, sym is surjective, $\ker(\text{sym})$ is α -invariant. Hence, there exists a natural $*$ -homomorphism

$$\pi \rtimes G : \Pi \rtimes_\alpha G \rightarrow ((C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})) \rtimes_\beta G.$$

We have an inclusion

$$((C_0 + \mathbb{C})(\mathbb{R}^N) \otimes_{\min} C(\mathbb{S}^{N-1})) \rtimes_\beta G \subset (L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})) \rtimes_\beta G.$$

Hence,

$$\pi \rtimes G : \Pi \rtimes_{\alpha} G \rightarrow (L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta} G.$$

Next, the standard trace $\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}}$ on $L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})$ is β -invariant. Hence, there exists a natural trace

$$\left(\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}} \right) \rtimes G$$

on $(L_{\infty}(\mathbb{R}^N) \bar{\otimes} L_{\infty}(\mathbb{S}^{N-1})) \rtimes_{\beta} G$.

2.5. Birman-Solomyak lemmas.

Lemma 2.7. *Let $A \in \mathcal{L}_{p,\infty}$ and $B \in (\mathcal{L}_{p,\infty})_0$. If*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{p}} \mu(t, A) = c,$$

then also

$$\lim_{t \rightarrow \infty} t^{\frac{1}{p}} \mu(t, A + B) = c.$$

Lemma 2.8. *Let $\{A_k\}_{k \geq 1} \subset \mathcal{L}_{p,\infty}$. If*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{p}} \mu(t, A_k) = c_k, \quad k \geq 1,$$

and if $A_k \rightarrow A$ in $\mathcal{L}_{p,\infty}$ as $k \rightarrow \infty$, then the following limits exist and are equal

$$\lim_{t \rightarrow \infty} t^{\frac{1}{p}} \mu(t, A) = \lim_{k \rightarrow \infty} c_k.$$

3. CHAMBER IS A DOUBLING METRIC MEASURE SPACE

Theorem 3.1. *Let Γ be a chamber. For all $x \in \Gamma$ and for all $r > 0$, we have*

$$w(B(x, r) \cap \Gamma) \geq c_{\kappa} w(B(x, r)).$$

In particular, $(\Gamma, |\cdot|, w(x)dx)$ is a doubling metric measure space.

Notation 3.2. *Let $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$. Set*

$$w_{\theta}(x) = \prod_{A \subset \{1, \dots, N_1\}} \left(\sum_{k \in A} x_k \right)^{\theta_A}, \quad x \in \mathbb{R}_+^{N_1}.$$

Lemma 3.3. *For every $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have*

$$\int_{B(x, r)} w_{\theta}(y_1, \dots, y_{N_1}) dy \leq c_{N, \theta}^{(1)} r^{N_1} w_{\theta}((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$B(x, r) \subset \left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r).$$

Thus,

$$\begin{aligned} & \int_{B(x, r)} w_{\theta}(y_1, \dots, y_{N_1}) dy \leq \\ & \leq \int_{\left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r)} w_{\theta}(y_1, \dots, y_{N_1}) dy = \\ & = \int_{\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r]} w_{\theta}(u) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \end{aligned}$$

$$\begin{aligned}
&\leq \int_{\times_{k=1}^{N_1} [0, x_k + r]} w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} = \\
&= w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times \prod_{k=1}^{N_1} (x_k + r - (x_k - r)_+) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\
&\leq w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (2r)^{N_1} \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2}.
\end{aligned}$$

□

Lemma 3.4. *For every $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have*

$$\int_{B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq c_{N, \theta}^{(2)} r^N w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}) \supset \left(\times_{k=1}^{N_1} \left[x_k + \frac{r}{3N_1}, x_k + \frac{r}{2N_1} \right] \right) \times B((x_{N_1+1}, \dots, x_N), \frac{r}{2}).$$

Thus,

$$\begin{aligned}
&\int_{B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq \\
&\geq \int_{\left(\times_{k=1}^{N_1} \left[x_k + \frac{r}{3N_1}, x_k + \frac{r}{2N_1} \right] \right) \times B((x_{N_1+1}, \dots, x_N), \frac{r}{2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq \\
&\geq \int_{\left(\times_{k=1}^{N_1} \left[x_k + \frac{r}{3N_1}, x_k + \frac{r}{2N_1} \right] \right) \times B((x_{N_1+1}, \dots, x_N), \frac{r}{2})} w_\theta(x_1 + \frac{r}{3N_1}, \dots, x_{N_1} + \frac{r}{3N_1}) dy = \\
&= \int_{\left(\times_{k=1}^{N_1} [0, \frac{r}{6N_1}] \right) \times B(0, \frac{r}{2})} w_\theta(x_1 + \frac{r}{3N_1}, \dots, x_{N_1} + \frac{r}{3N_1}) dy \geq \\
&\geq \int_{\left(\times_{k=1}^{N_1} [0, \frac{r}{6N_1}] \right) \times B(0, \frac{r}{2})} w_\theta(\frac{x_1 + r}{3N_1}, \dots, \frac{x_{N_1} + r}{3N_1}) dy = \\
&= w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times \left(\frac{r}{6N_1} \right)^{N_1} \times \text{Vol}(\mathbb{B}_{N_2}) \left(\frac{r}{2} \right)^{N_2} \times (3N_1)^{-\sum_{A \subset \{1, \dots, N_1\}} \theta_A}.
\end{aligned}$$

□

Lemma 3.5. *There exists $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ such that*

$$c_\kappa^{-1} w_\theta(x_1, \dots, x_{N_1}) \leq w(\Phi^{-1}(x)) \leq c_\kappa w_\theta(x_1, \dots, x_{N_1}), \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Proof. For every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ and for every $v \in R$, we have

$$\langle \Phi^{-1}(x), v \rangle = \langle x, (\Phi^{-1})^*(v) \rangle.$$

The latter expression has the same sign (positive or negative) on the whole cone $\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Hence, either $(\Phi^{-1})^*v \geq 0$ or $(\Phi^{-1})^*v \leq 0$ for every $v \in R$. In addition to that, $((\Phi^{-1})^*(v))_k = 0$ for $v \in R$ and for $k > N_1$. Let

$$A_v = \{1 \leq k \leq N_1 : ((\Phi^{-1})^*(v))_k \neq 0\}.$$

We, obviously, have

$$c_{\Phi, v}^{-1} \sum_{k \in A_v} x_k \leq |\langle \Phi^{-1}(x), v \rangle| \leq c_{\Phi, v} \sum_{k \in A_v} x_k, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Thus,

$$\prod_{v \in R} c_{\Phi, v}^{-\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)} \leq w(\Phi^{-1}(x)) \leq \prod_{v \in R} c_{\Phi, v}^{\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)}, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

This completes the proof. $\square$

Proof. For every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ and for every $v \in R$, we have

$$\langle \Phi^{-1}(x), v \rangle = \langle x, (\Phi^{-1})^*(v) \rangle.$$

The latter expression has the same sign (positive or negative) on the whole cone $\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Hence,

$$((\Phi^{-1})^*(v))_k = \begin{cases} \text{something positive,} & 1 \leq k \leq N_1 \\ 0, & k > N_1 \end{cases}$$

Let A_v be the set of those $1 \leq k \leq N_1$ with $((\Phi^{-1})^*(v))_l$ being strictly positive. We, obviously, have

$$c_{\Phi, v}^{-1} \sum_{k \in A_v} x_k \leq |\langle \Phi^{-1}(x), v \rangle| \leq c_{\Phi, v} \sum_{k \in A_v} x_k, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Thus,

$$\prod_{v \in R} c_{\Phi, v}^{-\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)} \leq w(x) \leq \prod_{v \in R} c_{\Phi, v}^{\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)}, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

This completes the proof. $\square$

Proof of Theorem 3.1. The first assertion follows by combining Lemmas 3.3, 3.4 and 3.5. The second assertion is an obvious consequence of the first one. $\square$

4. ROCHBERG-SEMMES BASED ESTIMATE

Theorem 4.1. *Let $N \geq 3$, $1 \leq N_1 \leq N$ and $N_2 = N - N_1$. Let $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Let measure $\mathbf{w}(x)dx$ be doubling on Ω and suppose that*

$$\sup_{x \in Q} \mathbf{w}(x) \leq c_{\mathbf{w}} \frac{\mathbf{w}(Q)}{m(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Suppose that $K : \Omega \times \Omega \rightarrow \mathbb{C}$ satisfies

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K, \alpha, \beta} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{\mathbf{w}(B(x, |x - y|))}, \quad x, y \in \Omega, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

Let V be an integral operator on $L_2(\Omega, \mathbf{w}(x)dx)$ with integral kernel K . For every $f \in C_c^\infty(\mathbb{R}^N)$, we have

$$\|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K, \mathbf{w}} \|f\|_{L_N(\Omega)}.$$

Lemma 4.2. *There exists a smooth function ψ supported on $[-1, 1]^N \setminus (-\frac{1}{4}, \frac{1}{4})^N$ such that for every $x, y \in \mathbb{R}^N$ with $x \neq y$,*

$$1 = \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that $\rho = 1$ on $(-\frac{1}{2}, \frac{1}{2})^N$ and $\rho = 0$ outside $(-1, 1)^N$. Set $\psi(z) = \rho(z) - \rho(2z)$. Clearly,

$$\text{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \frac{1}{4} \leq |z|_\infty \leq 1 \right\}.$$

For every $z \neq 0$, we have

$$(3) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

For every x , there exists a unique $Q \in \mathbb{D}_j$ containing x . For this Q , we have $\ell(Q) = 2^{-j}$. We have

$$\begin{aligned} 1 &= \sum_{j \in \mathbb{Z}} \psi(2^j(x-y)) = \sum_{j \in \mathbb{Z}} \left(\sum_{Q \in \mathbb{D}_j} \chi_Q(x) \right) \psi(2^j(x-y)) = \\ &= \sum_{j \in \mathbb{Z}} \sum_{Q \in \mathbb{D}_j} \chi_Q(x) \psi\left(\frac{x-y}{\ell(Q)}\right) = \sum_{Q \in \mathbb{D}} \chi_Q(x) \psi\left(\frac{x-y}{\ell(Q)}\right). \end{aligned}$$

If $x \in Q$ and $\psi(\frac{x-y}{\ell(Q)}) \neq 0$, then

$$x \in Q, \quad |x-y|_\infty \leq \ell(Q).$$

We obviously have $|x-c_Q|_\infty \leq \frac{1}{2}\ell(Q)$ and $|x-y|_\infty \leq \ell(Q)$. Thus, $|y-c_Q|_\infty \leq \frac{3}{2}\ell(Q)$. In other words, $y \in 3Q$. Thus,

$$\chi_Q(x) \psi\left(\frac{x-y}{\ell(Q)}\right) = \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x-y}{\ell(Q)}\right).$$

This completes the proof. $\square$

Lemma 4.3. *Let $r > 2$. In the setting of Theorem 4.1, we have*

$$f(x)K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k, l, Q}(x) f_{k, l, Q}(y), \quad x, y \in \Omega, \quad x \neq y,$$

where

- (i) $e_{k, l, Q}$ is supported in Q and $f_{k, l, Q}$ is supported in $3Q$ for every $Q \in \mathbb{D}$ with $Q \subset \Omega$;
- (ii) for every $k, l \in \mathbb{Z}^N$ and for every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have

$$\|e_{k, l, Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} = \|f\|_{L_r(Q, \mathbf{w}(x)dx)};$$

- (iii) for every $k, l \in \mathbb{Z}^N$ and for every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have

$$\|f_{k, l, Q}\|_{L_\infty(\mathbb{R}^N)} \leq \frac{c_{K, \theta} \cdot \ell(Q)}{\mathbf{w}(Q)};$$

Proof. For every $x, y \in \Omega$ with $x \neq y$, we have

$$\begin{aligned} K(x, y) &= \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) \\ &= \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right). \end{aligned}$$

For every $Q \in \mathbb{D}$ with $3Q \subset \Omega$, define the map

$$K_Q : [-\frac{1}{2}, \frac{1}{2}]^N \times [-\frac{3}{2}, \frac{3}{2}]^N \rightarrow \mathbb{C}$$

by setting

$$K_Q(a, b) = K(c_Q + al(Q), c_Q + bl(Q))\psi(a - b), \quad a \in [-\frac{1}{2}, \frac{1}{2}]^N, \quad b \in [-\frac{3}{2}, \frac{3}{2}]^N.$$

If $Q \in \mathbb{D}$ with $Q \subset \Omega$ is such that $3Q \not\subset \Omega$, then the definition of K_Q is the same, a also runs through $[-\frac{1}{2}, \frac{1}{2}]^N$, but b runs through a smaller parallelepiped. *To lighten the notation, in this proof, we write $[-\frac{3}{2}, \frac{3}{2}]^N$ instead of the domain of b .*

It follows from the assumption on K and from the doubling property of $\mathbf{w}(x)dx$ that that

$$\|K_Q\|_{C^{2N+2}([-\frac{1}{2}, \frac{1}{2}]^N \times [-\frac{3}{2}, \frac{3}{2}]^N)} \leq \frac{c'_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

Hence, there exists an extension $K_Q : \mathbb{T}^N \times \mathbb{T}^N \rightarrow \mathbb{C}$ such that

$$\|K_Q\|_{C^{2N+2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c''_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

In particular,

$$\|K_Q\|_{W^{2N+2,2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

We may, therefore, write

$$K_Q(a, b) = \sum_{k,l \in \mathbb{Z}^N} \text{coef}_{k,l,Q} e^{i\langle k,a \rangle} e^{i\langle l,b \rangle}, \quad a, b \in \mathbb{T}^N,$$

where

$$\sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{2N+2} |\text{coef}_{k,l,Q}|^2 \leq \left(\frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)} \right)^2.$$

In particular,

$$(1 + |k|^2 + |l|^2)^{N+1} \cdot |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}, \quad k, l \in \mathbb{Z}^N.$$

We now set

$$e_{k,l,Q}(x) = f(x)\chi_Q(x)e^{i\langle k, \frac{x-c_Q}{\ell(Q)} \rangle},$$

$$f_{k,l,Q}(y) = (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k,l,Q} \cdot \chi_{3Q}(y)e^{i\langle l, \frac{y-c_Q}{\ell(Q)} \rangle}.$$

This allows us to write

$$f(x)K(x, y) = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y).$$

Obviously,

$$\text{supp}(e_{k,l,Q}) \subset Q, \quad \text{supp}(f_{k,l,Q}) \subset 3Q,$$

$$\|e_{k,l,Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} = \|f\|_{L_r(Q, \mathbf{w}(x)dx)},$$

$$\|f_{k,l,Q}\|_{L_\infty(\mathbb{R}^N)} = (1 + |k|^2 + |l|^2)^{N+1} |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

□

Lemma 4.4. *Let $r > 2$. In the setting of Theorem 4.1, we have*

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K,\mathbf{w},r} \left\| \left\{ m(Q)^{\frac{1}{N}-\frac{1}{r}} \|f\|_{L_r(Q)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}.$$

Proof. Let $A_{k,l}$ be the integral operator with integral kernel given by the formula

$$(x, y) \rightarrow \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y), \quad x, y \in \Omega, \quad k, l \in \mathbb{Z}^N.$$

By Lemma 4.3, we have

$$M_f V = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k,l}.$$

By the triangle inequality,

$$\begin{aligned} & \|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq \\ & \leq \frac{N}{N-1} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))}. \end{aligned}$$

By Lemma 4.3, the sequences¹

$$\left\{ \mathbf{w}(Q)^{\frac{1}{r}-\frac{1}{2}} \frac{e_{k,l,Q}}{\|f\|_{L_r(Q, \mathbf{w}(x)dx)}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}}, \quad \left\{ \frac{c_{K,\theta}^{-1} \mathbf{w}(Q)^{\frac{1}{2}}}{\ell(Q)} f_{k,l,Q} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}}$$

are $(r, 1)$ -NWO and, respectively, $(\infty, 3)$ -NWO in the sense of Definition 2.4. Using Lemma 2.5, we conclude that

$$\begin{aligned} & \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq \\ & \leq c_{\theta,r,1,\infty,3,N} c_{K,\theta} \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}. \end{aligned}$$

Therefore,

$$\begin{aligned} & \|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq \\ & \leq \frac{N}{N-1} c_{\theta,r,1,\infty,3,N} c_{K,\theta} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \times \\ & \quad \times \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}. \end{aligned}$$

Obviously,

$$\mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \leq \left(\frac{1}{\mathbf{w}(Q)} \cdot \sup_{x \in Q} \mathbf{w}(x) \right)^{\frac{1}{r}} \times \|f\|_{L_r(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Combining the last two estimates, we complete the proof. $\square$

Lemma 4.5. *Let $1 < s < \infty$. We have*

$$\left\| \left\{ m(Q)^{\frac{1}{s}-1} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}} \right\|_{s,\infty} \leq 2s' \|F\|_{L_s(\mathbb{R}^N)}, \quad F \in L_s(\mathbb{R}^N).$$

¹The quotient in the first sequence is defined to be zero whenever $f = 0$ on Q .

Proof. Denote $s' = \frac{s}{s-1}$. For a finite set $E \subset \mathbb{D}$, denote

$$G_A = \sum_{Q \in E} m(Q)^{-\frac{1}{s'}} \chi_Q.$$

Since the dyadic cubes containing any fixed x are totally ordered by inclusion, for each $x \in \bigcup_{Q \in E} Q$ we may choose a cube $Q_x \in E$ of minimal volume such that $x \in Q_x$. Every $Q \in E$ containing x is, therefore, a dyadic ancestor of Q_x . Since the volume increases by the factor 2^N at each generation, it follows that

$$\begin{aligned} G_E(x) &= \sum_{\substack{Q \in E \\ x \in Q}} m(Q)^{-\frac{1}{s'}} \leq m(Q_x)^{-\frac{1}{s'}} \sum_{k=0}^{\infty} 2^{-\frac{kN}{s'}} = \frac{1}{1 - 2^{-\frac{N}{s'}}} m(Q_x)^{-\frac{1}{s'}} \leq \\ &\leq \frac{1}{1 - 2^{-\frac{1}{s'}}} m(Q_x)^{-\frac{1}{s'}} \leq 2s' m(Q_x)^{-\frac{1}{s'}} \leq 2s' \sup_{Q \in E} m(Q)^{-\frac{1}{s'}} \chi_Q(x). \end{aligned}$$

Consequently,

$$\begin{aligned} \|G_E\|_{L_{s'}(\mathbb{R}^N)}^{s'} &\leq (2s')^{s'} \int_{\mathbb{R}^N} \sup_{Q \in E} m(Q)^{-1} \chi_Q(x) dx \leq \\ &\leq (2s')^{s'} \int_{\mathbb{R}^N} \sum_{Q \in E} m(Q)^{-1} \chi_Q(x) dx = (2s')^{s'} |E|. \end{aligned}$$

Hence,

$$\|G_E\|_{L_{s'}(\mathbb{R}^N)} \leq 2s' |E|^{\frac{1}{s'}}.$$

By Hölder's inequality,

$$\sum_{Q \in E} \left(m(Q)^{-\frac{1}{s'}} \int_Q |F(x)| dx \right) = \int_{\mathbb{R}^N} |F(x)| G_E(x) dx \leq 2s' \|F\|_{L_s(\mathbb{R}^N)} \cdot |E|^{\frac{1}{s'}}.$$

In other words,

$$\sup_{\substack{E \subset \mathbb{D} \\ |E| < \infty}} |E|^{-\frac{1}{s'}} \left(\sum_{Q \in E} \left(m(Q)^{-\frac{1}{s'}} \int_Q |F(x)| dx \right) \right) \leq 2s' \|F\|_{L_s(\mathbb{R}^N)}.$$

Appealing to the definition of $l_{s,\infty}$, we complete the proof. $\square$

Proof of Theorem 4.1. Take $r \in (2, N)$. The assertion follows by combining Lemmas 4.4 and 4.5. $\square$

5. PROOF OF SPECIAL CBIKEL ESTIMATE IN DUNKL SETTING

Lemma 5.1. *For every pair of multi-indices $\alpha, \beta \in \mathbb{Z}_+^N$, for every $t > 0$, we have*

$$|(\partial_x^\alpha \partial_y^\beta h_t)(x, y)| \leq c_{\kappa, \alpha, \beta} t^{-\frac{|\alpha|_1 + |\beta|_1}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2).$$

Proof. This is (slightly weakened form of) Theorem 3.1 in [2]. $\square$

Lemma 5.2. *Suppose $s + N > 2$. For every $x, y \in \mathbb{R}^N$ with $d(x, y) > 0$, we have*

$$\int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq c_{\kappa, s} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s}.$$

Proof. By (2),

$$w(B(x, t^{\frac{1}{2}}))^{-1} \leq c_{\kappa}^{(1)} w(B(x, d(x, y)))^{-1} \times \begin{cases} \left(\frac{d(x, y)}{t^{\frac{1}{2}}}\right)^N, & 0 < t \leq d(x, y)^2, \\ \left(\frac{d(x, y)}{t^{\frac{1}{2}}}\right)^N, & t \geq d(x, y)^2. \end{cases}$$

Therefore,

$$\begin{aligned} & \int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp\left(-\frac{c_{\kappa}}{t} d(x, y)^2\right) dt \leq \\ & \leq c_{\kappa}^{(1)} w(B(x, d(x, y)))^{-1} \times \int_0^{d(x, y)^2} t^{-\frac{s}{2}} \left(\frac{d(x, y)}{t^{\frac{1}{2}}}\right)^N \exp\left(-\frac{c_{\kappa}}{t} d(x, y)^2\right) dt + \\ & \quad + c_{\kappa}^{(1)} w(B(x, d(x, y)))^{-1} \times \int_{d(x, y)^2}^\infty t^{-\frac{s}{2}} \left(\frac{d(x, y)}{t^{\frac{1}{2}}}\right)^N dt = \\ & = c_{\kappa}^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_0^1 t^{-\frac{s+N}{2}} \exp\left(-\frac{c_{\kappa}}{t}\right) dt + \\ & \quad + c_{\kappa}^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_1^\infty t^{-\frac{s+N}{2}} dt. \end{aligned}$$

□

The following lemma establishes derivative estimates for the integral kernels of Dunkl-Riesz transforms.

Lemma 5.3. *Let $N \geq 2$. For every pair of multi-indices $\alpha, \beta \in \mathbb{Z}_+^N$, we have*

$$|(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq c_{\kappa, \alpha, \beta} \frac{d(x, y)^{1-|\alpha|_1-|\beta|_1}}{w(B(x, d(x, y)))}, \quad x, y \in \mathbb{R}^N, \quad d(x, y) > 0.$$

Proof. We have

$$\Delta_{\text{Dunkl}}^{-\frac{1}{2}} = \pi^{-\frac{1}{2}} \int_0^\infty e^{-t \Delta_{\text{Dunkl}}} \frac{dt}{t^{\frac{1}{2}}}.$$

Hence,

$$K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(x, y) = \pi^{-\frac{1}{2}} \int_0^\infty h_t(x, y) \frac{dt}{t^{\frac{1}{2}}}.$$

Using Lemma 5.1, we estimate

$$\begin{aligned} & |(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq \\ & \leq \frac{c_{\kappa, \alpha, \beta}}{\pi^{\frac{1}{2}}} \int_0^\infty t^{-\frac{|\alpha|_1+|\beta|_1+1}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp\left(-\frac{c_{\kappa}}{t} d(x, y)^2\right) dt. \end{aligned}$$

Applying Lemma 5.2 with $s = 1 + |\alpha|_1 + |\beta|_1$, we conclude

$$\begin{aligned} & |(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq \\ & \leq \pi^{-\frac{1}{2}} c_{\kappa, \alpha, \beta} \times c_{\kappa, |\alpha|_1+|\beta|_1+1} w(B(x, d(x, y)))^{-1} \times d(x, y)^{1-|\alpha|_1-|\beta|_1}. \end{aligned}$$

The assertion follows immediately. □

Lemma 5.4. *For every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have*

$$\sup_{x \in Q} w_\theta(x) \leq c_\theta \frac{w_\theta(Q)}{m(Q)}.$$

Proof. Without loss of generality, we assume $\theta_\emptyset = 0$ and use the convention $0^0 = 1$. We have

$$w_\theta(Q) \geq c_\theta m(Q) \prod_{\emptyset \neq A \subset \{1, \dots, N_1\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

On the other hand, if $x \in Q$, then $|x - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$ and, therefore,

$$\sum_{k \in A} x_k \leq \frac{|A|}{2} \ell_Q + \sum_{k \in A} (c_Q)_k \leq N(\ell_Q + \sum_{k \in A} (c_Q)_k).$$

Thus,

$$\begin{aligned} w_\theta(x) &= \prod_{\emptyset \neq A \subset \{1, \dots, N_1\}} (\sum_{k \in A} x_k)^{\theta_A} \leq \\ &\leq N^{\sum_{\emptyset \neq A \subset \{1, \dots, N_1\}} \theta_A} \prod_{\emptyset \neq A \subset \{1, \dots, N_1\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}. \end{aligned}$$

This completes the proof. $\square$

Theorem 5.5. *Let $N \geq 3$. We have*

$$\|M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_\kappa \|f\|_{L_N(\mathbb{R}^N)}.$$

Proof. It suffices to estimate, for every chamber Γ and for every $g \in G$, the operator $M_{\chi_\Gamma \cdot f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_{\chi_{g\Gamma}}$. In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(x, y), \quad x \in \Gamma, \quad y \in g\Gamma,$$

on the Hilbert space $L_2(\mathbb{R}^N, w(x)dx)$. In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(x, gy), \quad x, y \in \Gamma,$$

on the Hilbert space $L_2(\Gamma, w(x)dx)$.

Let $N_1 = \dim(\text{span}(R))$, $N_2 = N - N_1$ and let $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. For a concrete chamber Γ , let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be a linear isomorphism given in Theorem 2.2 so that $\Phi(\Gamma) = \Omega$.

It suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x) K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space $L_2(\Omega, w(\Phi^{-1}x)dx)$. Using Lemma 3.5, it suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x) K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space $L_2(\Omega, w_\theta(x)dx)$.

By Lemma 5.3,

$$\left| \partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y) \right| \leq c_{\alpha, \beta, \kappa} \frac{|x - y|^{1 - |\alpha| - |\beta|_1}}{w_\theta(B(x, |x - y|))}, \quad x, y \in \Omega, \quad x \neq y.$$

The other assumptions in Theorem 4.1 follow from Lemma 5.4, Lemma 3.3 and Lemma 3.4. The assertion follows now from Theorem 4.1. $\square$

6. PROOF OF THEOREM 1.1 (i)

Lemma 6.1. *We have*

$$[T_{e_j}, M_f] = M_{\partial_j f} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} U_v,$$

where

$$(\delta_v f)(x) = \frac{f(x) - (U_v f)(x)}{\langle v, x \rangle}, \quad v \in R, \quad x \in \mathbb{R}^N.$$

Proof. Obvious. □

Lemma 6.2. *We have*

$$\|\delta_v f\|_{L_N(\mathbb{R}^N)} \leq c_N \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)}.$$

Proof. Without loss of generality, $v = e_N$. Set

$$(D_t h)(x) = h(x_1, \dots, x_{N-1}, tx_N).$$

We have

$$\delta_v f = \int_{-1}^1 D_t(\partial_N f) dt.$$

Thus,

$$\begin{aligned} \|\delta_v f\|_{L_N(\mathbb{R}^N)} &\leq \int_{-1}^1 \|D_t(\partial_N f)\|_{L_N(\mathbb{R}^N)} dt = \\ &= \int_{-1}^1 |t|^{-\frac{1}{N}} \|\partial_N f\|_{L_N(\mathbb{R}^N)} dt = \frac{2N}{N-1} \|\partial_N f\|_{L_N(\mathbb{R}^N)}. \end{aligned}$$

□

Lemma 6.3. *Let $N > 1$. Let B be a positive self-adjoint operator on a Hilbert space H with trivial kernel. Let $A \in B(H)$ satisfy $A : \text{dom}(B) \rightarrow \text{dom}(B)$. Suppose that $B^{-1}[B^2, A]B^{-1}$, interpreted in the quadratic-form sense on vectors with bounded spectral support away from zero, extends to an operator in $\mathcal{L}_{N,\infty}(H)$. We have*

$$\|[B, A]B^{-1}\|_{N,\infty} \leq c_N \|B^{-1}[B^2, A]B^{-1}\|_{N,\infty}.$$

Proof of Theorem 1.1 (i). We prove the assertion for $N \geq 3$ under *a priori* restriction $f \in C_c^\infty(\mathbb{R}^N)$. Passing to the case of $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ is standard (see e.g. [refer to our papers](#)).

We have

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] &= [T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} + T_{e_j} [\Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = \\ &= [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} - R_{\text{Dunkl},j} [\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \|[\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.3, we write

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq$$

$$\begin{aligned} &\leq \frac{N}{N-1} \| [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} c_N \| \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Note that

$$\begin{aligned} &\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} = \\ &= - \sum_{k=1}^N \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f] R_{\text{Dunkl},k} - \sum_{k=1}^N R_{\text{Dunkl},k} [T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} &\| \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ &\leq \frac{N}{N-1} \sum_{k=1}^N \| \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \sum_{k=1}^N \| [T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Combining the estimates in the last two paragraphs, we obtain

$$\begin{aligned} &\| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ &\leq c'_N \sum_{k=1}^N \| \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ c'_N \sum_{k=1}^N \| [T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.1, the fact that Δ_{Dunkl} commutes with U_v and the triangle inequality, we obtain

$$\begin{aligned} &\| [T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ &\leq \frac{N}{N-1} \| M_{\partial_k f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{2(N-1)} \sum_{v \in R} |v| \kappa(v) \| M_{\delta_v f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Thus,

$$\begin{aligned} &\| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ &\leq c''_N \sum_{k=1}^N \| M_{\partial_k f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ c_\kappa \sum_{v \in R} \| M_{\delta_v f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

The assertion follows now from Theorem 5.5 and Lemma 6.2. $\square$

7. PROOF OF THEOREM 1.1 (ii)

It is likely possible to combine a number of theorems in [9] (with Proposition 9.5 in [9] being the most important) in order to obtain this result. However, we propose a simpler direct proof. The advantage is twofold (a) we are able to use median method directly (b) proofs in [9] are not self-contained (Hytonen refers to GONG-LI-SINKO and, implicitly, to HYTONEN-2026), while our exposition is self-contained. [insert references.](#)

7.1. Abstract lower estimate.

Theorem 7.1. *Let $1 \leq j \leq N$. Let $K : \mathbb{R}^N \rightarrow \mathbb{R}$ be such that²*

$$K(x, y) \geq c_K \frac{y_j - x_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}, \quad x, y \in \mathbb{R}^N, \quad y_j \geq x_j.$$

Let $\mathbf{w} \in A_\infty$ and let V be an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel K . For every real-valued function $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ with $[V, M_f] \in \mathcal{L}_{N, \infty}(\mathbb{R}^N, \mathbf{w}(x)dx)$, we have $f \in \dot{W}^{1, N}(\mathbb{R}^N)$ and

$$\|f\|_{\dot{W}^{1, N}(\mathbb{R}^N)} \leq c_{K, \mathbf{w}} \| [V, M_f] \|_{\mathcal{L}_{N, \infty}(\mathbb{R}^N, \mathbf{w}(x)dx)}.$$

Lemma 7.2. *Let V be as in the assumptions of Theorem 7.1. Let real-valued $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ be such that the commutator $[V, M_f]$ is bounded on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$. For every $Q \in \mathbb{D}$, there exist ξ_Q and η_Q supported in $7Q$ such that*

$$\begin{aligned} \|\xi_Q\|_\infty, \|\eta_Q\|_\infty &\leq \mathbf{w}(Q)^{-\frac{1}{2}}, \\ \frac{1}{\mathbf{w}(Q)} \min_{c \in \mathbb{R}} \|f - c\|_{L_1(3Q, \mathbf{w}(x)dx)} &\leq c_{K, \mathbf{w}} |\langle [V, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)}|. \end{aligned}$$

Proof. It follows from the assumptions on K and from the doubling property of the measure $\mathbf{w}(x)dx$ that

$$K(x, y) \geq \frac{c_{K, \mathbf{w}}}{\mathbf{w}(Q)}, \quad x \in 3Q, \quad y \in Q + 3\ell(Q)e_j, \quad Q \in \mathbb{D}.$$

Here, $c_{K, \mathbf{w}}$ is some strictly positive constant.

Let α be the $\mathbf{w}(x)dx$ -median of f on $3Q$ and β be the $\mathbf{w}(x)dx$ -median of f on $Q + 3\ell(Q)e_j$. Assume for definiteness $\alpha \geq \beta$ (otherwise, replace f by $-f$). Set

$$\begin{aligned} A_1 &= \{x \in 3Q : f(x) \geq \alpha\}, \quad B_1 = \{x \in Q + 3\ell(Q)e_j : f(x) \leq \beta\}, \\ A_2 &= \{x \in 3Q : f(x) \leq \beta\}, \quad B_2 = \{x \in Q + 3\ell(Q)e_j : f(x) \geq \alpha\}, \\ \xi_{Q, k} &= \mathbf{w}(Q)^{-\frac{1}{2}} \chi_{B_k}, \quad \eta_{Q, k} = \mathbf{w}(Q)^{-\frac{1}{2}} \chi_{A_k}, \quad k = 1, 2. \end{aligned}$$

We have

$$\begin{aligned} \langle [V, M_f] \xi_{Q, k}, \eta_{Q, k} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} &= \\ &= \frac{1}{\mathbf{w}(Q)} \int_{A_k \times B_k} K(x, y) (f(y) - f(x)) \mathbf{w}(x) \mathbf{w}(y) dx dy. \end{aligned}$$

If $x \in A_1$ and $y \in B_1$, then

$$\begin{aligned} f(x) - f(y) &= (f(x) - \alpha) + (\alpha - \beta) + (\beta - f(y)) = \\ &= |f(x) - \alpha| + |\alpha - \beta| + |f(y) - \beta| \geq |f(x) - \alpha| + |\alpha - \beta| \geq 0. \end{aligned}$$

If $x \in A_2$ and $y \in B_2$, then

$$f(y) - f(x) = (f(y) - \beta) + (\beta - f(x)) = |f(y) - \beta| + |f(x) - \beta| \geq |f(x) - \beta| \geq 0.$$

²Here, c_K is some strictly positive constant.

Since $\mathbf{w} \in A_\infty(\mathbb{R}^N)$, it follows that the measure $\mathbf{w}(x)dx$ is doubling. By the definition of weighted medians,

$$\begin{aligned}\mathbf{w}(A_1) &\geq \frac{1}{2}\mathbf{w}(3Q) \geq \frac{1}{2}\mathbf{w}(Q), \\ \mathbf{w}(B_k) &\geq \frac{1}{2}\mathbf{w}(Q + 3\ell(Q)e_j) \geq c_{\mathbf{w}}\mathbf{w}(Q), \quad k = 1, 2.\end{aligned}$$

Thus,

$$\begin{aligned}& -\langle [V, M_f]\xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \geq \\ & \geq \frac{1}{\mathbf{w}(Q)} \int_{A_1 \times B_1} K(x, y) (|f(x) - \alpha| + |\alpha - \beta|) \mathbf{w}(x) \mathbf{w}(y) dx dy \geq \\ & \geq \frac{c_{K, \mathbf{w}}}{\mathbf{w}^2(Q)} \int_{A_1 \times B_1} (|f(x) - \alpha| + |\alpha - \beta|) \mathbf{w}(x) \mathbf{w}(y) dx dy = \\ & = \frac{c_{K, \mathbf{w}} \mathbf{w}(B_1)}{\mathbf{w}^2(Q)} \cdot \left(\int_{A_1} |f(x) - \alpha| \mathbf{w}(x) dx + |\alpha - \beta| \mathbf{w}(A_1) \right) \geq \\ & \geq \frac{c'_{K, \mathbf{w}}}{\mathbf{w}(Q)} \cdot \left(\int_{A_1} |f(x) - \alpha| \mathbf{w}(x) dx + |\alpha - \beta| \mathbf{w}(Q) \right)\end{aligned}$$

and

$$\begin{aligned}& \langle [V, M_f]\xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \geq \\ & \geq \frac{1}{\mathbf{w}(Q)} \int_{A_2 \times B_2} K(x, y) |f(x) - \beta| \mathbf{w}(x) \mathbf{w}(y) dx dy \geq \\ & \geq \frac{c_{K, \mathbf{w}}}{\mathbf{w}^2(Q)} \int_{A_2 \times B_2} |f(x) - \beta| \mathbf{w}(x) \mathbf{w}(y) dx dy = \\ & = \frac{c_{K, \mathbf{w}} \mathbf{w}(B_2)}{\mathbf{w}^2(Q)} \cdot \int_{A_2} |f(x) - \beta| \mathbf{w}(x) dx \geq \\ & \geq \frac{c'_{K, \mathbf{w}}}{\mathbf{w}(Q)} \int_{A_2} |f(x) - \beta| \mathbf{w}(x) dx.\end{aligned}$$

It remains to note that

$$\begin{aligned}\|f - \alpha\|_{L_1(3Q, \mathbf{w}(x)dx)} &= \int_{A_1} |f(y) - \alpha| \mathbf{w}(y) dy + \\ &+ \int_{A_2} |f(y) - \alpha| \mathbf{w}(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |f(y) - \alpha| \mathbf{w}(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| \mathbf{w}(y) dy + \int_{A_2} |f(y) - \beta| \mathbf{w}(y) dy + \\ &+ \int_{A_2} |\alpha - \beta| \mathbf{w}(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |\alpha - \beta| \mathbf{w}(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| \mathbf{w}(y) dy + \int_{A_2} |f(y) - \beta| \mathbf{w}(y) dy + |\alpha - \beta| \mathbf{w}(3Q).\end{aligned}$$

Since the measure $\mathbf{w}(x)dx$ is doubling, it follows that $\mathbf{w}(3Q) \leq c_{\mathbf{w}}\mathbf{w}(Q)$. Combining the last 3 inequalities, we conclude

$$\frac{c'_{K, \mathbf{w}}}{\mathbf{w}(Q)} \|f - \alpha\|_{L_1(3Q, \mathbf{w}(x)dx)} \leq \max_{k=1,2} |\langle [V, M_f]\xi_{Q,k}, \eta_{Q,k} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)}|.$$

If

$$|\langle [V, M_f]\xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)}| \geq |\langle [V, M_f]\xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)}|,$$

then we set $\xi_Q = \xi_{Q,1}$ and $\eta_Q = \eta_{Q,1}$. Otherwise, set $\xi_Q = \xi_{Q,2}$ and $\eta_Q = \eta_{Q,2}$. Thus,

$$\begin{aligned} \frac{c'_{K,\mathbf{w}}}{\mathbf{w}(Q)} \min_{c \in \mathbb{R}} \|f - c\|_{L_1(3Q, \mathbf{w}(x)dx)} &\leq \frac{c'_{K,\mathbf{w}}}{\mathbf{w}(Q)} \|f - \alpha\|_{L_1(3Q, \mathbf{w}(x)dx)} \leq \\ &\leq |\langle [V, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)}|. \end{aligned}$$

□

Lemma 7.3. *Let $\mathbf{w} \in A_p(\mathbb{R}^N)$ for $1 < p < \infty$. In the setting of Theorem 7.1, we have,*

$$\begin{aligned} &\left\| \left\{ \frac{1}{m(Q)^p} \min_{c \in \mathbb{R}} \|f - c\|_{L_{\frac{1}{p}}(3Q)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ &\leq c_{K,\mathbf{w},p} [\mathbf{w}]_{A_p(\mathbb{R}^N)} \|[V, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}. \end{aligned}$$

Proof. Let $\{\xi_Q\}_{Q \in \mathbb{D}}$ and $\{\eta_Q\}_{Q \in \mathbb{D}}$ be functions constructed in Lemma 7.2. By Lemma 2.6,

$$\begin{aligned} &\left\| \left\{ \langle [V, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ &\leq c_{\mathbf{w},\infty,7,\infty,7,N,\infty} \|[V, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}. \end{aligned}$$

By Lemma 7.2,

$$\begin{aligned} &\left\| \left\{ \frac{1}{\mathbf{w}(3Q)} \min_{c \in \mathbb{R}} \|f - c\|_{L_1(3Q, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ &\leq c_{K,\mathbf{w}} \|[V, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}. \end{aligned}$$

Applying Hölder inequality to the functions $|f - c|^{\frac{1}{p}} \mathbf{w}^{\frac{1}{p}}$ and $\mathbf{w}^{-\frac{1}{p}}$, we obtain

$$\int_{3Q} |f(x) - c|^{\frac{1}{p}} dx \leq \left(\int_{3Q} |f(x) - c| \mathbf{w}(x) dx \right)^{\frac{1}{p}} \left(\int_{3Q} \mathbf{w}^{-\frac{1}{p-1}}(x) dx \right)^{1-\frac{1}{p}}.$$

Taking the latter inequality to the power p , we obtain

$$\|f - c\|_{L_{\frac{1}{p}}(3Q)} \leq \|f - c\|_{L_1(3Q, \mathbf{w}(x)dx)} \cdot \left(\int_{3Q} \mathbf{w}^{-\frac{1}{p-1}}(x) dx \right)^{p-1}.$$

Thus,

$$\begin{aligned} &\left\| \left\{ \frac{1}{m(Q)^p} \min_{c \in \mathbb{R}} \|f - c\|_{L_{\frac{1}{p}}(3Q)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ &\leq c_{K,\mathbf{w}} \|[V, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))} \times \sup_{Q \in \mathbb{D}} \frac{\mathbf{w}(Q)}{m(Q)^p} \left(\int_{3Q} \mathbf{w}^{-\frac{1}{p-1}}(x) dx \right)^{p-1}. \end{aligned}$$

Since $\mathbf{w}(Q) \leq \mathbf{w}(3Q)$ and $m(3Q) = 3^N m(Q)$, it follows that

$$\begin{aligned} &\sup_{Q \in \mathbb{D}} \frac{\mathbf{w}(Q)}{m(Q)^p} \left(\int_{3Q} \mathbf{w}^{-\frac{1}{p-1}}(x) dx \right)^{p-1} \leq \\ &\leq 3^{Np} \sup_{Q \in \mathbb{D}} \frac{\mathbf{w}(3Q)}{m(3Q)^p} \left(\int_{3Q} \mathbf{w}^{-\frac{1}{p-1}}(x) dx \right)^{p-1} \leq 3^{Np} [\mathbf{w}]_{A_p(\mathbb{R}^N)}, \end{aligned}$$

where we used Definition 7.1.3 in [6]. This completes the proof. □

Lemma 7.4. *Let $N \geq 2$ and $f \in L_1^{\text{loc}}(\mathbb{R}^N)$. For every $p \in (1, \infty)$, we have*

$$\|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)} \leq c_{p,N} \left\| \left\{ \frac{1}{m(Q)^p} \min_{c \in \mathbb{R}} \|f - c\|_{L_{\frac{1}{p}}(3Q)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})}.$$

Proof. **supply reference!** □

Proof of Theorem 7.1. Let $p \in (1, \infty)$ be such that $\mathbf{w} \in A_p(\mathbb{R}^N)$. The assertion follows by combining Lemmas 7.3 and 7.4. $\square$

7.2. Proof of Theorem 1.1 (ii). Check the definition of Riesz transform and the sign in the below lemma.

Lemma 7.5. Let $1 \leq j \leq N$. Let $x, y \in \mathbb{R}^N$ be such that $d(x, y) > 0$. If $x_j \geq y_j$, then

$$K_{R_{\text{Dunkl},j}}(x, y) \leq c_\kappa \frac{y_j - x_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

If $x_j \leq y_j$, then

$$K_{R_{\text{Dunkl},j}}(x, y) \geq c_\kappa \frac{y_j - x_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

Proof. The proof is identical to that in [7, Theorem 1.2]. $\square$

Lemma 7.6. Let Q be a cube in $\mathbb{R}^N$. We have

$$\begin{aligned} \sup_{x \in Q} w(x) &\leq N^{\sum_{v \in R} \kappa(v)} \cdot \prod_{v \in R} (|\langle c_Q, v \rangle| + \ell(Q))^{\kappa(v)}, \\ \frac{w(Q)}{m(Q)} &\geq c_\kappa \prod_{v \in R} (|\langle c_Q, v \rangle| + \ell(Q))^{\kappa(v)}. \end{aligned}$$

Proof. If $x \in Q$, then $x = c_Q + y$, $|y| \leq N\ell(Q)$. Thus,

$$|\langle x, v \rangle| \leq |\langle c_Q, v \rangle| + |\langle y, v \rangle| \leq |\langle c_Q, v \rangle| + N\ell(Q).$$

Thus,

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{\kappa(v)} \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + N\ell(Q))^{\kappa(v)}.$$

This immediately yields first the assertion.

We have $B(c_Q, \frac{1}{2}\ell(Q)) \subset Q$. Thus,

$$\frac{w(Q)}{m(Q)} \geq \frac{w(B(c_Q, \frac{1}{2}\ell(Q)))}{m(Q)} = c_N \frac{w(B(c_Q, \frac{1}{2}\ell(Q)))}{m(B(c_Q, \frac{1}{2}\ell(Q)))}.$$

The second assertion follows now from (1). $\square$

Lemma 7.7. We have $w \in A_\infty(\mathbb{R}^N)$. Thus, $w \in A_p(\mathbb{R}^N)$ for some $1 < p < \infty$.

Proof. It follows from Lemma 7.6

$$\sup_{x \in Q} w(x) \leq \frac{c_\kappa}{m(Q)} \int_Q w(x) dx$$

holds for every cube in $\mathbb{R}^N$. The assertion follows from Theorem 7.3.3 (c) in [6]. $\square$

Proof of Theorem 1.1 (ii). Since $R_{\text{Dunkl},j}$ is skew-adjoint on $L_2(\mathbb{R}^N, w(x)dx)$, it follows that

$$[R_{\text{Dunkl},j}, M_f]^* = [R_{\text{Dunkl},j}, M_{\bar{f}}].$$

Thus,

$$[R_{\text{Dunkl},j}, M_{\Re(f)}], [R_{\text{Dunkl},j}, M_{\Im(f)}] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Hence, we may assume without loss of generality that f is real-valued. It follows from Lemmas 7.5 and 7.7 that the assumptions in Theorem 7.1 are satisfied for $V = R_{\text{Dunkl},j}$ and $\mathbf{w} = w$. The assertion follows now from Theorem 7.1. $\square$

8. PROOF OF SMOOTHNESS

Theorem 8.1. *Let $f \in C_c^\infty(\mathbb{R}^N)$.*

(i) *for every $n \in \mathbb{N}$, $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 = n$, the operator³*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

is bounded on $L_2(\mathbb{R}^N, w(x)dx)$;

(ii) *for every $n \in \mathbb{N}$, $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 > n$, the operator⁴*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$;

Lemma 8.2. *Let $f \in C_c^\infty(\mathbb{R}^N)$. We have*

$$[\Delta_{\text{Dunkl}}, M_f] = -2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} - \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} (T_{e_j} U_v + U_v T_{e_j}) + \sum_{g \in G} M_{f_g} U_g.$$

All functions on the right hand side are smooth and compactly supported.

Proof. We have

$$\begin{aligned} [\Delta_{\text{Dunkl}}, M_f] &= - \sum_{j=1}^N (T_{e_j} [T_{e_j}, M_f] + [T_{e_j}, M_f] T_{e_j}) = \\ &= - \sum_{j=1}^N (T_{e_j} M_{\partial_j f} + M_{\partial_j f} T_{e_j}) - \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (T_{e_j} M_{\delta_v f} U_v + M_{\delta_v f} U_v T_{e_j}) = \\ &= -2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} - \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} U_v + M_{\delta_v f} U_v T_{e_j}) + \\ &\quad - \sum_{j=1}^N [T_{e_j}, M_{\partial_j f}] - \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) [T_{e_j}, M_{\delta_v f}] U_v = \\ &= -2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} - \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} U_v + M_{\delta_v f} U_v T_{e_j}) + \\ &\quad - \sum_{j=1}^N M_{\partial_j^2 f} - \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\delta_v (\partial_j f)} U_v + \\ &\quad - \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\partial_j (\delta_v f)} U_v - \frac{1}{4} \sum_{j=1}^N \sum_{v, v' \in R} v_j v'_j \kappa(v) \kappa(v') M_{\delta_{v'} \delta_v f} U_{v'} U_v. \end{aligned}$$

Absorbing the signs of the zero-order terms into the functions f_g , we obtain the assertion. $\square$

³Commutators are repeated n times

⁴Commutators are repeated n times

Lemma 8.3. *Let $N \geq 3$. For every $f \in C_c^\infty(\mathbb{R}^N)$, the operator*

$$[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

is bounded and

$$(1 + \Delta_{\text{Dunkl}})^{-1}[\Delta_{\text{Dunkl}}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We prove only the first assertion. The proof of the second one is similar (modulo Theorem 5.5).

Since each T_{e_j} commutes with $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$, each U_v commutes with $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$ and since the operators

$$\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, \quad 1 \leq j \leq N,$$

are bounded, the assertion follows from Lemma 8.2. $\square$

Lemma 8.4. *Let $N \geq 3$ and $f \in C_c^\infty(\mathbb{R}^N)$. For every $n \in \mathbb{N}$, the operator⁵*

$$[\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\dots, M_f] \dots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}$$

is bounded.

Proof. Denote for brevity

$$\text{Expr}_n(f) = [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\dots, M_f] \dots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}.$$

We prove the assertion by induction on f . The n -th statement is "for every $f \in C_c^\infty(\mathbb{R}^N)$, the operator $\text{Expr}_n(f)$ is bounded." Base of induction is given by Lemma 8.3. It suffices to prove the step of induction.

Since each T_{e_j} and each U_v commutes with Δ_{Dunkl} , it follows from Lemma 8.2 that

$$\begin{aligned} \text{Expr}_{n+1}(f) &= -2 \sum_{j=1}^N \text{Expr}_n(\partial_j f) \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} - \\ &\quad \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) \text{Expr}_n(\delta_v f) \cdot \left(\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} U_v + U_v \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \right) + \\ &\quad + \sum_{g \in G} \text{Expr}_n(f_g) \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_g. \end{aligned}$$

In each summand, the first factor is bounded by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof. $\square$

Proof of Theorem 8.1 (i). The assertion follows from Lemma 8.4 and Hadamard 3 lines theorem. $\square$

Lemma 8.5. *Let $N \geq 3$. For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-1} \in \mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

⁵Commutators are repeated n times

Proof. Set

$$\mathcal{D} = \sum_{j=1}^N \gamma_j \otimes T_{e_j},$$

where $\{\gamma_j\}_{j=1}^N$ are matrices on $\mathbb{C}^{2^{\lfloor N/2 \rfloor}}$ satisfying $\gamma_j^* = -\gamma_j$ and $\gamma_j \gamma_k + \gamma_k \gamma_j = -2\delta_{jk}I$. We have

$$\begin{aligned} 1 \otimes M_f(1 + \Delta_{\text{Dunkl}})^{-1} &= (1 \otimes M_f)(\mathcal{D}^2 + 1)^{-1} = (1 \otimes M_f)(\mathcal{D} + i)^{-1}(\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + [1 \otimes M_f, (\mathcal{D} + i)^{-1}](\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1}[\mathcal{D}, 1 \otimes M_f](\mathcal{D}^2 + 1)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1}\left(\sum_{j=1}^N \gamma_j \otimes [T_{e_j}, M_f]\right)(\mathcal{D}^2 + 1)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1}(\gamma_j \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1} + \\ &\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (\mathcal{D} + i)^{-1}(\gamma_j \otimes M_{\delta_v f} U_v)(\mathcal{D}^2 + 1)^{-1}. \end{aligned}$$

Since Δ_{Dunkl} commutes with U_v , it follows that

$$\begin{aligned} 1 \otimes M_f(1 + \Delta_{\text{Dunkl}})^{-1} &= \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes 1) + \\ &\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (\mathcal{D} + i)^{-1}(1 \otimes M_{\delta_v f})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes U_v). \end{aligned}$$

Clearly,

$$\begin{aligned} &\|(\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} = \\ &= 2^{\frac{2\lfloor N/2 \rfloor}{N}} \|(\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}} M_f (\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ &\leq 2^{\frac{2\lfloor N/2 \rfloor}{N}} \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ &\leq 2^{\frac{2\lfloor N/2 \rfloor}{N}} 2^{\frac{2}{N}} \|M_{|f|^{\frac{1}{2}}} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))}^2 \leq \\ &\stackrel{Th. 5.5}{\leq} c_{N, \kappa} \| |f|^{\frac{1}{2}} \|_{L_N(\mathbb{R}^N)}^2 = c_{N, \kappa} \|f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)} \end{aligned}$$

and, similarly,

$$\begin{aligned} &\|(\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|\partial_j f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}, \\ &\|(\mathcal{D} + i)^{-1}(1 \otimes M_{\delta_v f})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|\delta_v f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}. \end{aligned}$$

This completes the proof. $\square$

Lemma 8.6. *Let $N \geq 3$. For every integer $n \geq 1$ and every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We prove the assertion by induction on f . The n -th statement is "for every $f \in C_c^\infty(\mathbb{R}^N)$, we have

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))."$$

Base of induction (i.e. cases $n = 1$ and $n = 2$) are given by Theorem 5.5 and Lemma 8.5. It suffices to prove the step of induction.

We have

$$\begin{aligned} & M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + [M_f, (1 + \Delta_{\text{Dunkl}})^{-1}](1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}}. \end{aligned}$$

Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $(f \circ U_g) \cdot \phi = f \circ U_g$ for every $g \in G$. It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

Thus,

$$\begin{aligned} & M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} = (1 + \Delta_{\text{Dunkl}})^{-1} M_f \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + \\ &+ (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}. \end{aligned}$$

In the first summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.5 and the second one belongs to the ideal $\mathcal{L}_{\frac{N}{n-1}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption. In the second summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.3, the second factor belongs to the ideal $\mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption and the third one is obviously bounded. $\square$

Lemma 8.7. *Let $N \geq 3$. Let $k, l \in \mathbb{Z}_+$ with $l > k$. For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$.

Proof. We prove the assertion by induction on k . Base of induction (i.e., the case $k = 0$) is established in Lemma 8.6.

We have

$$\begin{aligned} & T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} = T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f T_{e_{m_k}}(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} [T_{e_{m_k}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} = \\ &= T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times \frac{T_{e_{m_k}}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\partial_{m_k} f}(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \\ &+ \frac{1}{2} \sum_{v \in R} v_{m_k} \kappa(v) T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\delta_v f}(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_v. \end{aligned}$$

In each summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof. $\square$

Lemma 8.8. *Let $N \geq 3$ and $f \in C_c^\infty(\mathbb{R}^N)$. Let $s_1, s_2 \in \mathbb{R}$ be such that $s_1 + s_2 > 0$. We have*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$.

Proof. Step 1: Let us prove the assertion for $s_1, s_2 \in \mathbb{Z}$.

Suppose first $s_1 \leq 0$. Denote $k = -s_1 \geq 0$ and $l = s_2 > k$. Denote for brevity

$$V = (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} (1 + \sum_{j_1, \dots, j_k=1}^N |T_{e_{j_1}} \cdots T_{e_{j_k}}|)^{-1}.$$

Obviously, V is bounded. We have

$$\begin{aligned} (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} &= V \times (1 + \sum_{m_1, \dots, m_k=1}^N |T_{e_{m_1}} \cdots T_{e_{m_k}}|) = V + \\ &+ \sum_{m_1, \dots, m_k=1}^N V \times \text{sgn}(T_{e_{m_1}} \cdots T_{e_{m_k}})^* \times T_{e_{m_1}} \cdots T_{e_{m_k}}. \end{aligned}$$

It, therefore, suffices to prove that operators

$$M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}, \quad T_{e_{m_1}} \cdots T_{e_{m_k}} M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belong to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. This is established in Lemma 8.6 and Lemma 8.7, respectively. Hence, the claim is proved for $s_1, s_2 \in \mathbb{Z}$ and $s_1 \leq 0$.

If $s_2 \leq 0$, then the claim follows from that for $s_1 \leq 0$ by taking the adjoint. If $s_1, s_2 \geq 0$, then, according to Lemma 8.7 the operators

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}} M_f, \quad M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}}$$

belong to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. The claim for $s_1, s_2 \geq 0$ follows now from the Hadamard 3 lines theorem.

Step 2: Set $A = 1 + \Delta_{\text{Dunkl}}$. For every integer $m \geq 0$, we have

$$A^m M_f A^{-m} = \sum_{j=0}^m \binom{m}{j} \underbrace{[\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, \dots, [\Delta_{\text{Dunkl}}, M_f] \cdots]]}_{j \text{ commutators}} A^{-j},$$

where the summand corresponding to $j = 0$ is M_f . By Lemma 8.4, every summand on the right hand side is bounded. Thus,

$$A^m M_f A^{-m} \in B(L_2(\mathbb{R}^N, w(x)dx)), \quad m \geq 0.$$

Taking adjoints and applying the Hadamard 3 lines theorem, we obtain

$$A^t M_f A^{-t} \in B(L_2(\mathbb{R}^N, w(x)dx)), \quad t \in \mathbb{R}.$$

Consequently, for every $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 = 0$, we have

$$A^{-\frac{s_1}{2}} M_f A^{-\frac{s_2}{2}} \in B(L_2(\mathbb{R}^N, w(x)dx)).$$

On the other hand, Step 1 and complex interpolation imply that, for every integer $m \geq 1$ and every $s_1 \in \mathbb{R}$,

$$A^{-\frac{s_1}{2}} M_f A^{-\frac{m-s_1}{2}} \in \mathcal{L}_{\frac{N}{m}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Indeed, this follows by interpolating between two consecutive integer values of s_1 while keeping $s_1 + s_2 = m$.

Finally, interpolating between two consecutive nonnegative integer values of $s_1 + s_2$, using the preceding boundedness estimate when $s_1 + s_2 = 0$, yields

$$A^{-\frac{s_1}{2}} M_f A^{-\frac{s_2}{2}} \in \mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx)), \quad s_1 + s_2 > 0.$$

□

Proof of Theorem 8.1 (ii). Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $(f \circ U_g) \cdot \phi = f \circ U_g$ for every $g \in G$. It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

We have

$$\begin{aligned} & (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{n-s_1}{2}} \times \\ & \quad \times (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1-n}{2}} M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}. \end{aligned}$$

The first factor is bounded by Theorem 8.1 (i). The second factor belongs to $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.8. □

9. APPROXIMATION THEOREM

Theorem 9.1. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} M_{w^{\frac{1}{2}}} [R_{\text{Dunkl}, j}, M_f] M_{w^{-\frac{1}{2}}} &\in V \Delta^{-\frac{1}{2}} + (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N)), \\ V &= M_{\partial_j f} + \frac{1}{2} \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} U_v - \sum_{k=1}^N R_j M_{\partial_k f} R_k - \\ &\quad - \frac{1}{4} R_j \cdot \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} (R_k U_v + U_v R_k). \end{aligned}$$

9.1. First approximation theorem.

Theorem 9.2. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} & [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{1}{2} [\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-1} \in \\ & \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Lemma 9.3. *Let $B \geq 1$. [Need the priori assumption on A and B here] Suppose*

$$B^{-1} [B^2, [B^2, A]] B^{-\frac{5}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

We have

$$[B, A] B^{-1} - \frac{1}{2} [B^2, A] B^{-2} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

Proof. We have

$$B = \frac{2}{\pi} \int_0^\infty \frac{B^2 d\lambda}{\lambda^2 + B^2}.$$

Hence,

$$[B, A]B^{-1} = \frac{2}{\pi} \int_0^\infty \left[\frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} d\lambda.$$

Obviously,

$$\begin{aligned} \left[\frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} &= - \left[\frac{\lambda^2}{\lambda^2 + B^2}, A \right] B^{-1} = \frac{\lambda^2}{\lambda^2 + B^2} [B^2, A] \frac{1}{B(\lambda^2 + B^2)} = \\ &= [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} - \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2}. \end{aligned}$$

Obviously,

$$\begin{aligned} \int_0^\infty [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} d\lambda &= [B^2, A] B^{-1} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + B^2)^2} d\lambda = \\ &= [B^2, A] B^{-2} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + 1)^2} d\lambda = \frac{\pi}{4} [B^2, A] B^{-2}. \end{aligned}$$

On the other hand,

$$\begin{aligned} &\left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ &\leq \left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \|B^{-1} [B^2, [B^2, A]] B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty} \left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty. \end{aligned}$$

Next,

$$\left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{\lambda^2 t}{\lambda^2 + t^2} \leq \lambda$$

and

$$\left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{t^{\frac{3}{2}}}{(\lambda^2 + t^2)^2} \leq (\lambda^2 + 1)^{-\frac{5}{4}}.$$

Thus,

$$\begin{aligned} &\left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ &\leq \frac{\lambda}{(\lambda^2 + 1)^{\frac{5}{4}}} \|B^{-1} [B^2, [B^2, A]] B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty}. \end{aligned}$$

Hence, the integral

$$\int_0^\infty \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} d\lambda$$

converges absolutely in $\mathcal{L}_{\frac{2N}{3}, \infty}$ and the assertion follows. $\square$

Proof of Theorem 9.2. The assertion follows from Theorem 8.1 and Lemma 9.3. $\square$

9.2. Second approximation theorem.

Theorem 9.4. *If $f \in C_c^\infty(\mathbb{R}^N)$, then*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Lemma 9.5. *Let A, B_1, B_2 be such that $B_1, B_2 \geq 1$. Assume that B_1, B_2 are positive self-adjoint operators on a Hilbert space H , that $A \in B(H)$ satisfies $A(\text{Dom}(B_1)) \subset \text{Dom}(B_1)$, and that, for every $\lambda \geq 0$,*

$$\frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} = \frac{1}{\lambda^2 + B_1}(B_2 - B_1)\frac{1}{\lambda^2 + B_2}$$

holds in the sense of bounded extensions. Suppose that

- (i) $A(B_2 - B_1)B_2^{-\frac{1}{2}}$, $[B_1, A]B_1^{-\frac{1}{2}}$ and $[B_1, A]B_2^{-\frac{1}{2}}$ are bounded;
- (ii) *there exists a bounded operator A' with $B_1^{-\frac{3}{4}}A' \in \mathcal{L}_{\frac{2N}{3},\infty}$ such that $A = A'A$ and $[B_1, A] = A'[B_1, A]$;*

We have

$$AB_1^{-\frac{1}{2}} - AB_2^{-\frac{1}{2}} \in \mathcal{L}_{\frac{2N}{3},\infty}.$$

Proof. We have

$$B_k^{-\frac{1}{2}} = \frac{2}{\pi} \int_0^\infty \frac{d\lambda}{\lambda^2 + B_k}, \quad k = 1, 2.$$

Thus,

$$AB_1^{-\frac{1}{2}} - AB_2^{-\frac{1}{2}} = \frac{2}{\pi} \int_0^\infty A \left(\frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} \right) d\lambda.$$

Obviously,

$$\begin{aligned} A \left(\frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} \right) &= A \frac{1}{\lambda^2 + B_1} (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} + \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_1} (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} + \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_2}. \end{aligned}$$

Consider, for example, the first summand. We write

$$\begin{aligned} \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} &= \frac{1}{\lambda^2 + B_1} A' \cdot A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{B_1^{\frac{3}{4}}}{\lambda^2 + B_1} \cdot B_1^{-\frac{3}{4}} A' \cdot A(B_2 - B_1) B_2^{-\frac{1}{2}} \cdot \frac{B_2^{\frac{1}{2}}}{\lambda^2 + B_2}. \end{aligned}$$

Thus,

$$\begin{aligned} &\left\| \frac{1}{\lambda^2 + B_1} A(B_2 - B_1) \frac{1}{\lambda^2 + B_2} \right\|_{\frac{2N}{3},\infty} \leq \\ &\leq \left\| \frac{B_1^{\frac{3}{4}}}{\lambda^2 + B_1} \right\|_\infty \cdot \|B_1^{-\frac{3}{4}} A'\|_{\frac{2N}{3},\infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot \left\| \frac{B_2^{\frac{1}{2}}}{\lambda^2 + B_2} \right\|_\infty \leq \\ &\leq \sup_{t \geq 1} \frac{t^{\frac{3}{4}}}{\lambda^2 + t} \cdot \|B_1^{-\frac{3}{4}} A'\|_{\frac{2N}{3},\infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot \sup_{t \geq 1} \frac{t^{\frac{1}{2}}}{\lambda^2 + t} \leq \\ &\leq (\lambda^2 + 1)^{-\frac{1}{4}} \cdot \|B_1^{-\frac{3}{4}} A'\|_{\frac{2N}{3},\infty} \cdot \|A(B_2 - B_1) B_2^{-\frac{1}{2}}\|_\infty \cdot (\lambda^2 + 1)^{-\frac{1}{2}}. \end{aligned}$$

Hence, the first summand is absolutely integrable in $\mathcal{L}_{\frac{2N}{3}, \infty}$. Similarly, so are the other summands. This completes the proof. $\square$

Lemma 9.6. *Let P be a differential operator of degree 1 with smooth compactly supported coefficients. Suppose that the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$. The operators*

$$P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}, \quad P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

are bounded on $L_2(\mathbb{R}^N, w(x)dx)$.

Proof. By assumption, $P = M_{w^{-\frac{1}{2}}} P' M_{w^{\frac{1}{2}}}$, where P' is a differential operator of degree 1 with smooth compactly supported coefficients. Thus,

$$P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} = M_{w^{-\frac{1}{2}}} \cdot P'(1 + \Delta)^{-\frac{1}{2}} \cdot M_{w^{\frac{1}{2}}}.$$

Boundedness of the right hand side is obvious.

Similarly,

$$P = \sum_{j=1}^N M_{f_j} T_{e_j} + \sum_{g \in G} M_{h_g} U_g,$$

where functions $\{f_j\}_{j=1}^N$ and $\{h_g\}_{g \in G}$ are smooth and compactly supported. Since the operators $\{U_g\}_{g \in G}$ commute with Δ_{Dunkl} , it follows that

$$P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} = \sum_{j=1}^N M_{f_j} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \sum_{g \in G} M_{h_g} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} U_g.$$

Boundedness of the right hand side is obvious. $\square$

Proof of Theorem 9.4. Step 1: Suppose first that f is supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$.

We apply Lemma 9.5 with $A = M_f$, $B_1 = 1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}$ and $B_2 = 1 + \Delta_{\text{Dunkl}}$.

Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $f = f\phi$ and set $A' = M_\phi$. We have

$$M_\phi[M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f] = [M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f].$$

Thus, $A = A'A$ and $[B_1, A] = A'[B_1, A]$. It follows from the usual Cwikel estimates that $B_1^{-\frac{3}{4}} A' \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. This verifies the assumption (ii) in Lemma 9.5.

Note that $[B_1, A]$ is a differential operator P of degree 1 with smooth compactly supported coefficients. Furthermore, the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$. By Lemma 9.6, the operators

$$[B_1, A]B_1^{-\frac{1}{2}} = P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \quad [B_1, A]B_2^{-\frac{1}{2}} = P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}},$$

are bounded on $L_2(\mathbb{R}^N, w(x)dx)$. Note that there exist differential operators $\{P_g\}_{g \in G}$ of degree 1 with smooth compactly supported coefficients such that the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$ and such that

$$A(B_2 - B_1) = \sum_{g \in G} P_g U_g.$$

Since $\{U_g\}_{g \in G}$ commute with Δ_{Dunkl} , it follows that

$$A(B_2 - B_1) = \sum_{g \in G} P_g (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \cdot U_g$$

is bounded on $L_2(\mathbb{R}^N, w(x)dx)$. This verifies the assumption (i) in Lemma 9.5.

Using Lemma 9.5, we infer the claim of Step 1.

Step 2: Choose a sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ such that (a) $0 \leq \phi_n \leq 1$ for $n \geq 1$ (b) ϕ_n is supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}$, $v \in R$, for $n \geq 1$ (c) $\text{supp}(\phi_n) \uparrow \mathbb{R}^N$ as $n \rightarrow \infty$.

By Step 1,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$$

for every $n \geq 1$.

By Theorem 5.5,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \rightarrow M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ as $n \rightarrow \infty$. By the usual Cwikel estimate,

$$M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \rightarrow M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ as $n \rightarrow \infty$. Thus

$$\begin{aligned} \lim_{n \rightarrow \infty} M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} &= \\ &= M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \end{aligned}$$

where the limit is understood in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Since the separable part $(\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$ is closed in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, the assertion follows. $\square$

9.3. Proof of Theorem 9.1.

Lemma 9.7. *We have*

$$[R_{\text{Dunkl},j}, M_f] - [\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] \in \mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Set

$$h(x) = 1 - \frac{|x|}{(1 + |x|^2)^{\frac{1}{2}}}, \quad x \in \mathbb{R}^N.$$

We, obviously, have

$$R_{\text{Dunkl},j} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} = R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}).$$

Thus,

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] - [\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] &= \\ &= R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) M_f - M_f h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j} = \\ &= R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) (1 + \Delta_{\text{Dunkl}}) \cdot (1 + \Delta_{\text{Dunkl}})^{-1} M_f - \\ &\quad - M_f (1 + \Delta_{\text{Dunkl}})^{-1} \cdot (1 + \Delta_{\text{Dunkl}}) h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j}. \end{aligned}$$

In the first summand on the right hand side, the first and the second factors are bounded and the third one belongs to $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.6. In the second summand on the right hand side, the second and the third factors are bounded and the first one belongs to $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.6. $\square$

Lemma 9.8. *Let $\{B_j\}_{j=1}^N$ and $\{D_j\}_{j=1}^N$ be commuting tuples of self-adjoint operators on $L_2((\mathbb{R})^N, w(x)dx)$. Denote*

$$\langle B \rangle = (1 + \sum_{j=1}^N B_j^2)^{\frac{1}{2}}, \quad \langle D \rangle = (1 + \sum_{j=1}^N D_j^2)^{\frac{1}{2}}.$$

Suppose that there exists a sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ with

- (i) $0 \leq \phi_n \leq 1$ for $n \geq 1$ and $\text{supp}(\phi_n) \uparrow \mathbb{R}^N$ as $n \rightarrow \infty$;
- (ii)

$$M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)), \quad n \geq 1;$$

- (iii) $M_{\phi_n}(B_k - D_k)$ is bounded⁶ and compactly supported for $n \geq 1$ and $1 \leq k \leq N$;
- (iv) $[D_k, M_{f\phi_n}]$ is bounded⁷ and compactly supported for $n \geq 1$ and $1 \leq k \leq N$;
- (v) $[\langle D \rangle^{-1}, M_{f\phi_n}] \in (\mathcal{L}_{N,\infty})_0(\mathbb{R}^N, w(x)dx)$ for every $n \geq 1$;
- (vi)

$$\|M_h \langle B \rangle^{-1}\|_{\mathcal{L}_{N,\infty}(\mathbb{R}^N, w(x)dx)} \leq c_N \|h\|_{L_N(\mathbb{R}^N)}, \quad h \in L_N(\mathbb{R}^N),$$

$$\|M_h \langle D \rangle^{-1}\|_{\mathcal{L}_{N,\infty}(\mathbb{R}^N, w(x)dx)} \leq c_N \|h\|_{L_N(\mathbb{R}^N)}, \quad h \in L_N(\mathbb{R}^N);$$

We have

$$\frac{B_j}{\langle B \rangle} M_f \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Choose real-valued $\psi_n \in C_c^\infty(\mathbb{R}^N)$ such that

$$\begin{aligned} M_{\phi_n}(B_k - D_k) &= M_{\phi_n}(B_k - D_k) \cdot M_{\psi_n}, \quad 1 \leq k \leq N, \\ [M_{f\phi_n}, D_k] &= M_{\psi_n} \cdot [M_{f\phi_n}, D_k] \cdot M_{\psi_n}, \quad 1 \leq k \leq N. \end{aligned}$$

We have

$$\begin{aligned} & \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} = \frac{B_j}{\langle B \rangle} \cdot M_f \left(M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1} \right) \cdot \frac{B_k}{\langle B \rangle} + \\ & + \frac{B_j}{\langle B \rangle} \cdot [M_{f\phi_n}, \langle D \rangle^{-1}] \cdot \frac{B_k}{\langle B \rangle} + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_f \cdot M_{\phi_n}(B_k - D_k) \cdot M_{\psi_n} \langle B \rangle^{-1} + \\ & + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_{\psi_n} \cdot [M_{f\phi_n}, D_k] \cdot M_{\psi_n} \langle B \rangle^{-1} + \frac{B_j}{\langle B \rangle} \frac{D_k}{\langle D \rangle} \cdot M_f \left(M_{\phi_n} \langle B \rangle^{-1} - M_{\phi_n} \langle D \rangle^{-1} \right) + \\ & + \frac{B_j}{\langle B \rangle} \cdot \langle D \rangle^{-1} M_{\psi_n} \cdot [D_k, M_{f\phi_n}] \cdot M_{\psi_n} \langle D \rangle^{-1} + \frac{B_j}{\langle B \rangle} \cdot [\langle D \rangle^{-1}, M_{f\phi_n}] \cdot \frac{D_k}{\langle D \rangle} + \\ & + \langle B \rangle^{-1} M_{\psi_n} \cdot (B_j - D_j) M_{\phi_n} \cdot M_f \frac{D_k}{\langle D \rangle^2} + \\ & + \langle B \rangle^{-1} M_{\psi_n} \cdot [D_j, M_{f\phi_n}] \cdot M_{\psi_n} \frac{D_k}{\langle D \rangle^2} + (\langle B \rangle^{-1} - \langle D \rangle^{-1}) M_{\phi_n} \cdot M_f \frac{D_j D_k}{\langle D \rangle^2} + \\ & + \langle D \rangle^{-1} M_{\psi_n} \cdot [M_{f\phi_n}, D_j] \cdot M_{\psi_n} \frac{D_k}{\langle D \rangle^2}. \end{aligned}$$

It follows now from the assumptions and from Hölder inequality that

$$\frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

⁶By a constant which depends also on n .

⁷By a constant which depends also on n .

Obviously,

$$\begin{aligned}
& \left\| \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{B_j}{\langle B \rangle} M_f \frac{B_k}{\langle B \rangle^2} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
& \leq \left\| M_{f(1-\phi_n)} \langle B \rangle^{-1} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_N \|f(1-\phi_n)\|_{L_N(\mathbb{R}^N)}, \\
& \left\| \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\
& \leq \left\| M_{f(1-\phi_n)} \langle D \rangle^{-1} \right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_N \|f(1-\phi_n)\|_{L_N(\mathbb{R}^N)}.
\end{aligned}$$

Thus,

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \frac{B_j}{\langle B \rangle} M_{f\phi_n} \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_{f\phi_n} \frac{D_k}{\langle D \rangle^2} = \\
& = \frac{B_j}{\langle B \rangle} M_f \frac{B_k}{\langle B \rangle^2} - \frac{D_j}{\langle D \rangle} M_f \frac{D_k}{\langle D \rangle^2},
\end{aligned}$$

where the limit is taken in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Since $(\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$ is closed in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, the assertion follows. $\square$

Lemma 9.9. *For every $f \in C_c^\infty(\mathbb{R}^N)$,*

$$\begin{aligned}
& \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} M_f \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v \in \\
& \in M_{w^{-\frac{1}{2}}} R_j M_f R_k U_v \Delta^{-\frac{1}{2}} M_{w^{\frac{1}{2}}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)), \\
& \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} M_f U_v \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} \in \\
& \in M_{w^{-\frac{1}{2}}} R_j M_f U_v R_k \Delta^{-\frac{1}{2}} M_{w^{\frac{1}{2}}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

Proof. The assertion follows by applying Lemma 9.8 to the tuples $(T_{e_1}, \dots, T_{e_N})$ and $(M_{w^{-\frac{1}{2}}} \partial_1 M_{w^{\frac{1}{2}}}, \dots, M_{w^{-\frac{1}{2}}} \partial_N M_{w^{\frac{1}{2}}})$ and sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}_{v \in R}$. $\square$

Proof of Theorem 9.1. We have

$$\begin{aligned}
& \left[\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f \right] = \\
& = [T_{e_j}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}.
\end{aligned}$$

Using Lemma 9.7 and Theorem 9.2, we write

$$\begin{aligned}
& [R_{\text{Dunkl},j}, M_f] \in [T_{e_j}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \\
& - \frac{1}{2} \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-1} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

It follows from Lemma 8.2 and Lemma 8.6 that

$$\begin{aligned}
& [\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-1} - 2 \sum_{k=1}^N M_{\partial_k f} \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} - \\
& - \frac{1}{2} \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} \left(\frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v + U_v \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} \right) \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

Thus,

$$\begin{aligned}
[R_{\text{Dunkl},j}, M_f] &\in M_{\partial_j f}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} U_v - \\
&\quad - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot \sum_{k=1}^N M_{\partial_k f} \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} - \\
&\quad - \frac{1}{4} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} \left(\frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} U_v + U_v \frac{T_{e_k}}{1 + \Delta_{\text{Dunkl}}} \right) \in \\
&\quad \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

The assertion follows from Lemma 9.9. $\square$

10. SPECTRAL ASYMPTOTIC FORMULA

Theorem 10.1. *Let $N > 2$. Let Π be the C^* -algebra of operators admitting principal symbol. For every $T \in \Pi \rtimes_\alpha G$ compactly supported from the right, we have*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, T \Delta^{-\frac{1}{2}}) = c_N \|(\text{sym} \rtimes G)(T)\|_{L_N((L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})) \rtimes_\beta G)}.$$

Definition 10.2. *Let G be a Weyl group, acting orthogonally on $\mathbb{R}^N$, and let Γ be a fixed chamber. Let $\{e_g\}_{g \in G}$ be the standard basis of $\mathbb{C}^G$. Define a surjective partial isometry $W : L_2(\mathbb{R}^N) \otimes \mathbb{C}^G \rightarrow L_2(\mathbb{R}^N)$ by setting*

$$W \left(\sum_{g \in G} \xi_g \otimes e_g \right) = \sum_{g \in G} (\xi_g|_\Gamma) \circ g^{-1}.$$

The crucial observation is

$$\mu(W^* T W) = \mu(T), \quad T \in B(L_2(\mathbb{R}^N)).$$

Indeed, this is true for every surjective isometry.

Lemma 10.3. *If $f \in C_c^\infty(\mathbb{R}^N)$, then*

$$M_f \Delta^{-\frac{1}{2}} \in \sum_{g \in G} M_{f \chi_{g\Gamma}} \Delta^{-\frac{1}{2}} M_{\chi_{g\Gamma}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N)).$$

Proof. This follows from Theorem 4.1 in [5]. $\square$

Lemma 10.4. *If $f \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$ and if $h \in C^\infty(\mathbb{S}^{N-1})$, then*

$$h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f \in \sum_{g \in G} M_{\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f \chi_{g\Gamma}} + \mathcal{K}(L_2(\mathbb{R}^N)).$$

Proof. Indeed, $f \chi_{g\Gamma} \in C_c^\infty(\mathbb{R}^N)$ for $g \in \Gamma$. Hence, **this requires a reference!!**

$$[h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f \chi_{g\Gamma}}] \in \mathcal{K}(L_2(\mathbb{R}^N)), \quad g \in G.$$

The assertion follows now from the following algebraic identity.

$$h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f = \sum_{g \in G} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f \chi_{g\Gamma}} = \sum_{g \in G} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f \chi_{g\Gamma}} \cdot M_{\chi_{g\Gamma}} =$$

$$\begin{aligned}
&= \sum_{g \in G} M_{f\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) \cdot M_{\chi_{g\Gamma}} + \sum_{g \in G} [h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f\chi_{g\Gamma}}] \cdot M_{\chi_{g\Gamma}} = \\
&= \sum_{g \in G} M_{\chi_{g\Gamma}} \cdot M_{f\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) \cdot M_{\chi_{g\Gamma}} + \sum_{g \in G} [h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f\chi_{g\Gamma}}] \cdot M_{\chi_{g\Gamma}} = \\
&= \sum_{g \in G} M_{\chi_{g\Gamma}} \cdot h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{f\chi_{g\Gamma}} \cdot M_{\chi_{g\Gamma}} + \sum_{g \in G} M_{\chi_{g\Gamma}} \cdot [M_{f\chi_{g\Gamma}}, h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right)] \cdot M_{\chi_{g\Gamma}} + \\
&\quad + \sum_{g \in G} [h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right), M_{f\chi_{g\Gamma}}] \cdot M_{\chi_{g\Gamma}}.
\end{aligned}$$

□

Lemma 10.5. *Let $f \in C_c^\infty(\mathbb{R}^N)$ and let $g \in G$. We have*

$$\begin{aligned}
W^* M_f W &= \sum_{g \in G} M_{(f\chi_{g\Gamma}) \circ g} \otimes E_{g,g}, \\
W^* (M_{\chi_{g\Gamma}} \Delta^{-\frac{1}{2}} M_{\chi_{g\Gamma}}) W &= M_{\chi_\Gamma} \Delta^{-\frac{1}{2}} M_{\chi_\Gamma} \otimes E_{g,g}, \\
W^* M_{\chi_{g\Gamma}} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{\chi_{g\Gamma}} W &= M_{\chi_\Gamma} (h \circ g) \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{\chi_\Gamma} \otimes E_{g,g}, \\
W^* U_g W &= \sum_{g' \in G} M_{\chi_\Gamma} \otimes E_{gg',g'}.
\end{aligned}$$

Proof. The third assertion follows from the equality

$$U_g^{-1} h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) U_g = (h \circ g) \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right).$$

The other assertions are obvious. □

Lemma 10.6. *If $f \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$ and if $h \in C^\infty(\mathbb{S}^{N-1})$, then*

$$\begin{aligned}
&W^* h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f \Delta^{-\frac{1}{2}} W \in \\
&\in \sum_{g \in G} \left[M_{\chi_\Gamma} (h \circ g) \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{(f\chi_{g\Gamma}) \circ g} \Delta^{-\frac{1}{2}} M_{\chi_\Gamma} \right] \otimes E_{g,g} + (\mathcal{L}_{N,\infty})_0 (L_2(\mathbb{R}^N) \otimes \mathbb{C}^G).
\end{aligned}$$

Proof. We write $f = f_1 f_2$, where $f_1, f_2 \in C_c^\infty(\mathbb{R}^N)$ are supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$. The assertion follows from Lemma 10.3 (applied with f_2 in place of f), Lemma 10.4 (applied with f_1 in place of f) and Lemma 10.5. □

Lemma 10.7. *If $f \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$ and if $h \in C^\infty(\mathbb{S}^{N-1})$, then*

$$\begin{aligned}
&W^* h\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_f \Delta^{-\frac{1}{2}} U_g W \in \\
&\in \sum_{g' \in G} \left[(h \circ g') \left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right) M_{(f\chi_{g'\Gamma}) \circ g'} \Delta^{-\frac{1}{2}} \right] \otimes E_{g',g^{-1}g'} + (\mathcal{L}_{N,\infty})_0 (L_2(\mathbb{R}^N) \otimes \mathbb{C}^G).
\end{aligned}$$

Proof. The assertion follows from Lemma 10.6 and the facts (see the proof of Lemma 10.4 for the first one and Theorem 4.1 in [5] for the second one) that

$$M_{\chi_\Gamma}(h \circ g)\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right)M_F - (h \circ g)\left(\frac{\nabla}{\Delta^{\frac{1}{2}}}\right)M_F \in \mathcal{K}(L_2(\mathbb{R}^N)),$$

$$M_F\Delta^{-\frac{1}{2}}M_{\chi_\Gamma} - M_{F\chi_\Gamma}\Delta^{-\frac{1}{2}} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N)),$$

for every $F \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$. $\square$

Theorem 10.8. *Let $N > 2$. Let Π be the C^* -algebra of operators admitting principal symbol. For every $S \in \Pi \otimes M_m(\mathbb{C})$, compactly supported from the right, we have*

$$\lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, S(\Delta^{-\frac{1}{2}} \otimes 1)) = c_N \|(\text{sym} \otimes \text{id})(S)\|_{L_N(L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1}) \otimes M_m(\mathbb{C}))}.$$

Lemma 10.9. *Let $A = L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})$. Recall that G acts on A by the formula*

$$(\beta(g))(f \otimes 1) = (f \circ g^{-1}) \otimes 1, \quad (\beta(g))(1 \otimes h) = 1 \otimes (h \circ g^{-1}).$$

There is $$ -homomorphism $\Phi : A \rtimes_\beta G \rightarrow A \otimes M_G(\mathbb{C})$ such that*

$$\Phi(aU_g) = \sum_{g' \in G} (\beta(g'^{-1}))(a) \otimes E_{g', g^{-1}g'}, \quad a \in A, \quad g \in G.$$

We have

$$\left(\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}} \otimes \text{Tr} \right) \circ \Phi = |G| \cdot \left(\left(\int_{\mathbb{R}^N} \otimes \int_{\mathbb{S}^{N-1}} \right) \rtimes_\beta G \right).$$

the proof is available for convenience of co-authors. however, it is too elementary to be included into the text. so, it is commented out.

Lemma 10.10. *Let $X \in A \rtimes_\beta G$. Let*

$$V_g = \sum_{g' \in G} E_{g'g^{-1}, g} \in M_G(\mathbb{C})$$

be a unitary matrix. We have

$$(\beta(g) \otimes \text{id})(\Phi(X)) = (1 \otimes 1 \otimes V_g^{-1})\Phi(X)(1 \otimes 1 \otimes V_g).$$

the proof is available for convenience of co-authors. however, it is too elementary to be included into the text. so, it is commented out.

Proof of Theorem 10.1. Step 1: Suppose first

$$T = \sum_{g \in G} \sum_{k=1}^K h_{k,g} \left(\frac{\nabla}{\Delta^{\frac{1}{2}}} \right) M_{f_{k,g}} U_g$$

with $h_{k,g} \in C^\infty(\mathbb{S}^{N-1})$ and with $f_{k,g} \in C_c^\infty(\mathbb{R}^N)$ supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$.

Using Lemma 10.7, we write

$$W^* T \Delta^{-\frac{1}{2}} W \in S(\Delta^{-\frac{1}{2}} \otimes 1) + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N) \otimes \mathbb{C}^G),$$

where

$$S = \sum_{g, g' \in G} \sum_{k=1}^K \left[(h_{k,g} \circ g') \left(\frac{\nabla}{\Delta^{\frac{1}{2}}} \right) M_{(f_{k,g} \chi_{g'\Gamma}) \circ g'} \right] \otimes E_{g', g^{-1}g'}.$$

Obviously, $S \in \Pi \otimes M_G(\mathbb{C})$ is compactly supported from the right. We have

$$\begin{aligned} (\text{sym} \otimes \text{id})(S) &= \sum_{g, g' \in G} \sum_{k=1}^K \left[((f_{k,g} \chi_{g'\Gamma}) \circ g') \otimes (h_{k,g} \circ g') \right] \otimes E_{g', g^{-1}g'} = \\ &\stackrel{L.10.9}{=} \Phi \left(\sum_{g \in G} \sum_{k=1}^K (f_{k,g} \otimes h_{k,g}) \right) \cdot (\chi_\Gamma \otimes 1_{C(\mathbb{S}^{N-1})} \otimes 1_{M_G(\mathbb{C})}). \end{aligned}$$

Using Theorem 10.8 and Lemma 2.7, we conclude that

$$\begin{aligned} \lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, T \Delta^{-\frac{1}{2}}) &= \lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, W^* T \Delta^{-\frac{1}{2}} W) = \lim_{t \rightarrow \infty} t^{\frac{1}{N}} \mu(t, S(\Delta^{-\frac{1}{2}} \otimes 1)) = \\ &= c_N \left\| \Phi \left(\sum_{g \in G} \sum_{k=1}^K (f_{k,g} \otimes h_{k,g}) \right) \cdot (\chi_\Gamma \otimes 1_{C(\mathbb{S}^{N-1})} \otimes 1_{M_G(\mathbb{C})}) \right\|_{L_N(L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1}) \otimes M_G(\mathbb{C}))}. \end{aligned}$$

For every $X \in L_N((L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})) \rtimes_\beta G)$, we have

$$\begin{aligned} &\left\| \Phi(X) \cdot (\chi_\Gamma \otimes 1_{C(\mathbb{S}^{N-1})} \otimes 1_{M_G(\mathbb{C})}) \right\|_{L_N(L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1}) \otimes M_G(\mathbb{C}))} = \\ &\stackrel{L.10.10}{=} |G|^{-\frac{1}{N}} \left\| \Phi(X) \right\|_{L_N(L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1}) \otimes M_G(\mathbb{C}))} = \\ &\stackrel{L.10.9}{=} \|X\|_{L_N((L_\infty(\mathbb{R}^N) \bar{\otimes} L_\infty(\mathbb{S}^{N-1})) \rtimes_\beta G)}. \end{aligned}$$

This completes the proof for our special T .

Step 2: Consider now the general T . There exists (see e.g. the proof of Theorem 3.5 in [?]) a sequence $T_K \rightarrow T$ in $\Pi \rtimes_\alpha G$ such that

$$T_K = T'_K + \sum_{g \in G} \sum_{k=1}^K h_{K,k,g} \left(\frac{\nabla}{\Delta^{\frac{1}{2}}} \right) M_{f_{K,k,g}} U_g, \quad 1 \leq k \leq K.$$

Here, $h_{K,k,g} \in C^\infty(\mathbb{S}^{N-1})$, $f_{K,k,g} \in C_c^\infty(\mathbb{R}^N)$ are supported away from the hyperplanes $\{\langle x, v \rangle = 0\}_{v \in R}$, $T'_K \in \mathcal{K}(L_2(\mathbb{R}^N))$ is compactly supported from the right. The assertion follows now from Step 1 and Lemmas 2.7, 2.8. $\square$

REFERENCES

- [1] Nicolas Bourbaki. *Lie groups and Lie algebras. Chapters 4–6*. Elements of Mathematics (Berlin). Springer-Verlag, Berlin, 2002. Translated from the 1968 French original by Andrew Pressley. [3](#)
- [2] Jacek Dziubański and Agnieszka Hejna. Remark on atomic decompositions for the Hardy space H^1 in the rational Dunkl setting. *Studia Math.*, 251(1):89–110, 2020. [11](#)
- [3] Z. Fan, M. Lacey, and J. Li. Schatten classes and commutators of Riesz transform on Heisenberg group and applications. *J. Fourier Anal. Appl.*, 29(2):Paper No. 17, 28, 2023. [3](#)
- [4] Z. Fan, M. Lacey, J. Li, M.N. Vempati, and B.D. Wick. Besov space, Schatten classes, and commutators of Riesz transforms associated with the Neumann Laplacian. *Michigan Math. J.*, 75(1):199–224, 2025. [3](#)
- [5] R.L. Frank, F. Sukochev, and D. Zanin. Asymptotics of singular values for quantum derivatives. *Trans. Amer. Math. Soc.*, 376(3):2047–2088, 2023. [32](#), [34](#)
- [6] L. Grafakos. *Classical Fourier analysis*, volume 249 of *Graduate Texts in Mathematics*. Springer, New York, third edition, 2014. [18](#), [19](#)
- [7] Y. Han, M. Lee, J. Li, and B.D. Wick. Riesz transforms and commutators in the Dunkl setting. *Anal. Math. Phys.*, 14(3):Paper No. 46, 32, 2024. [19](#)
- [8] S. Helgason. *Differential geometry, Lie groups, and symmetric spaces*, volume 80 of *Pure and Applied Mathematics*. Academic Press, Inc. [Harcourt Brace Jovanovich, Publishers], New York-London, 1978. [2](#)

- [9] T. Hytonen. Schatten properties of commutators on metric spaces. Preprint available at <https://arxiv.org/abs/2411.02613>, 2024. [3](#), [4](#), [16](#)
- [10] M.T. Lacey, J. Li, and B.D. Wick. Schatten classes and commutator in the two weight setting, I. Hilbert transform. *Potential Anal.*, 60(2):875–894, 2024. [3](#)
- [11] J. Li, X. Xiong, and F. Yang. Schatten properties of Calderón–Zygmund singular integral commutator on stratified Lie groups. *J. Math. Pures Appl. (9)*, 188:73–113, 2024. [3](#)
- [12] R. Rochberg and S. Semmes. Nearly weakly orthonormal sequences, singular value estimates, and Calderon-Zygmund operators. *J. Funct. Anal.*, 86(2):237–306, 1989. [3](#)
- [13] Z. Wei and H. Zhang. Complex median method and Schatten class membership of commutators. Preprint available at <https://arxiv.org/abs/2411.05810>, 2024. [3](#)
- [14] Z. Wei and H. Zhang. On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators. Preprint available at <https://arxiv.org/abs/2407.14988>, 2024. [3](#)

ZHIJIE FAN, SCHOOL OF MATHEMATICAL SCIENCES, SOUTH CHINA NORMAL UNIVERSITY, GUANGZHOU 510631, CHINA

Email address: `fanzhj3@mail2.sysu.edu.cn`

JI LI, DEPARTMENT OF MATHEMATICS, MACQUARIE UNIVERSITY, NSW2109, AUSTRALIA

Email address: `ji.li@mq.edu.au`

SCHOOL OF MATHEMATICS AND STATISTICS, HNP-LAMA, CENTRAL SOUTH UNIVERSITY, CHANGSHA 410075, CHINA

Email address: `d.zanin@csu.edu.cn`